Unit 2: Data Transmission

1. Differentiate between [8] 2014 [F]

   i. Synchronous and Asynchronous Communication
   ii. Serial and Parallel Transmission

2. Write short notes on:

   i. RS-232 Interface Standard [5] 2014 [F]
   ii. Line Configuration [5] 2014 [S]
   iii. Shannon Capacity theorem [5] 2025[S](New)

3. Discuss different types of serial data transmission. Also illustrate a situation when baud rate can be lower than bit rate. [7] 2015 [F]

4. A TV channel has a signal with bandwidth of 12 MHz & it is transmitted with the average power of 6W. This signal has to travel through a channel which introduces an average noise of 0.1 mW. If the TV signal is digitized and sent, find maximum data rate of the channel.

5.Explain the differences between synchronous and asynchronous serial communication. [5] 2015 [S]

6.Explain different serial communication mode. [4] 2016 [F]

7.Define Baud Rate. Explain RS-232C standards. [7] 2016 [F]

8. Differentiate parallel and serial transmission. Explain different modes of serial transmission. [4+4] 2017 [F],2021 [S]

9. Explain null-modem configuration in RS-232C.Explain the relationship between baud rate and bit rate. [5+3] 2014 [F], 2021 [S]

10. Explain maximum channel capacity of noiseless and noisy channel. [7] 2017 [F]

11. Define Baud rate. An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need? [7] 2018 [S]

12. On a transmission channel, 600 channels message using 7-bit code is used. For sync data stream two sync characters and single errors detection is added. In asynchronous data transmission there is one start bit, one stop bit and error detection character is added. Calculate the efficiency of transmission in both transmission modes. [7] 2018 [F]

13. Define Bit rate and Baud rate. Explain the RS-232C Standard with handshake signals. [8] 2019 [F]

14. Explain data frame format of synchronous and asynchronous communications and differentiate between them. [7] 2019 [S]

15. What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. [7] 2020 [F]

16. Explain the concept of bit rate, baud rate. Explain data frame format of synchronous and asynchronous communication. [8] 2021[F]

17. Differentiate between different types of transmission mode. Briefly explain RS-232C. [8] 2022 [F]

18. Define bit rate and baud rate. Explain with a scenario to show it rate is not always equal to baud rate with suitable diagrams and calculation. [7] 2023 [S]

Sol:

Case 1 for two level of voltage i.e. 0 and 1

Each signal element has 2 levels → 1 bit per signal

i.e. Each signal change = 1 bit

Calculation

Baud rate (r)= 1000 signals/sec

Bits per signal (n)= $\mathbf{\log_2(2) = 1}$ **or** $\mathbf{L=2^n}$

$$\mathbf{2^1=2^n}$$

n=1 (bits per signal)

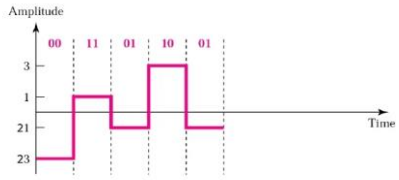
Bit rate (R) = r*n

R=1000*1=1000 bits per sec

Bit rate = Baud rate

Case 2 for 4 levels of voltage: Multilevel Signaling :00 ,01 ,10 ,11

Each signal represents 2 bits now



Calculation

Baud rate (r)= 1000 signals/sec

Bits per signal (n)= $\mathbf{\log_2(4) = 2}$ **or** $\mathbf{L=2^n}$

$$\mathbf{2^4=2^n}$$

n=2 (bits per signal)

Bit rate (R) = r*n

R=1000*2=2000 bits per sec (bps)

Result

Bit rate = 2000 bps

Baud rate = 1000 baud

Bit rate >Baud rate

19. Write short notes on [5] 2023 [S]

- Synchronous and Asynchronous Communication

20. What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. [7] 2024 [S] (old)

21. Explain Shannon's theorem for a noisy channel. Find the maximum bit rate for channel having bandwidth of 3100 Hz and a signal-to-noise ratio (SNR) of 20 dB. [7] 2024 [S] (New)

22. Define Baud rate. An analog signal has a bit rate of 800 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need? [7] 2024 [F] (old)

23.What is Serial and Parallel data transmission? Differentiate between synchronous and asynchronous data transmission. [8] 2024 [F] (New)

24.Discuss about serial data communication. A system sends a signal that can assume 8 different voltage levels. It sends 400 of these signals per second. What are the baud and bit rates? [8] 2025 [S]

Sol:

Number of signal levels (L) = 8

Baud rate (r)= number of signal units transmitted per second.

Baud rate (r) = 400 baud

$L=2^n$ **0r bits per signal=$\log_2 L$**

$8=2^n$ **or bits per signal = $\log_2 8= \log_2 2^3 =3$**

$2^3=2^n$

$n=3$

we know,

$$n=\frac{R}{r}$$

$$R=n*r$$

$$=3*400$$

$$=1200 \text{ bits/sec(bps)}$$

the baud rate(r)=400 baud

and bit rates(R)=1200 bps